

A Differentiable Physics-Prior Residual 3D U-Net for Control-Point-Level Photon Dose Prediction on CT (DoseRAD2026 Photon-CT Task)

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Abstract. We predict the Monte-Carlo (MC) dose of every photon MLC control point with an analytical, differentiable physics prior followed by a residual 3D U-Net. Six channels (density, radiological depth, primary fluence, distance-to-CAX, source distance, and a naive analytical dose surrogate) are computed per control point from the CT density and the beam descriptor; a 3D U-Net (dilated bottleneck, Softplus head) learns only the residual to MC. Because the aperture enters as a fluence *input*, the model is robust to the benchmark’s deliberate shift from random training apertures to clinical VMAT test apertures. In ablation, the analytical prior contributes $+1.9\gamma$ (1%/1 mm) and hard-region loss weighting $+2.8$, stacking to $+4.0$; multi-modal joint training across real- and synthetic-CT inputs adds a further $+2.8$ (cross-validated, every fold). Under 5-fold cross-validation on the 75 public patients the model reaches plan γ 1%/1 mm of 93.4 ± 1.0 (multi-modal variant 96.2 ± 1.1). The final submission is trained on all 75 patients and deployed as a knowledge-distilled base-32 student ($2.1\times$ faster, *above* the teacher’s held-out gamma) with aperture-bbox cropping, GPU-resident channel construction, compiled inference and cutoff-zeroed output: on the hidden preliminary test set it scored $\gamma=95.33$, per-beam MAE 0.0107, IDD 0.0072, stratified MAE 0.0049, DVH 0.30 at 80.3 s per job, well within the 181 s gate.

Keywords: photon dose prediction · differentiable physics · residual learning · knowledge distillation · VMAT

1 Introduction

Fast, accurate photon dose calculation underpins online adaptive radiotherapy: a plan must be recomputed on the anatomy of the day within minutes, at Monte-Carlo (MC)-level accuracy. Deep dose engines approach this [1,2], but most treat transport as an implicit target the network must memorize end-to-end. We instead factor the problem: an analytical, *differentiable* physics operator $\mathcal{C}$ maps the CT density and the control-point descriptor to dose-shaped input channels, and a light 3D U-Net $\mathcal{D}_\theta$ learns only the residual between this prior and

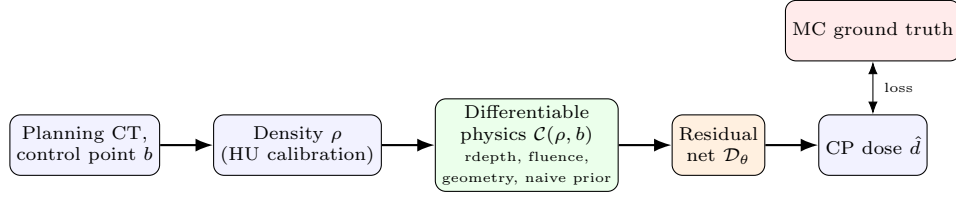


Fig. 1. Photon-CT pipeline: per control point, differentiable physics channels from the CT density; a 3D U-Net corrects the analytical prior toward MC.

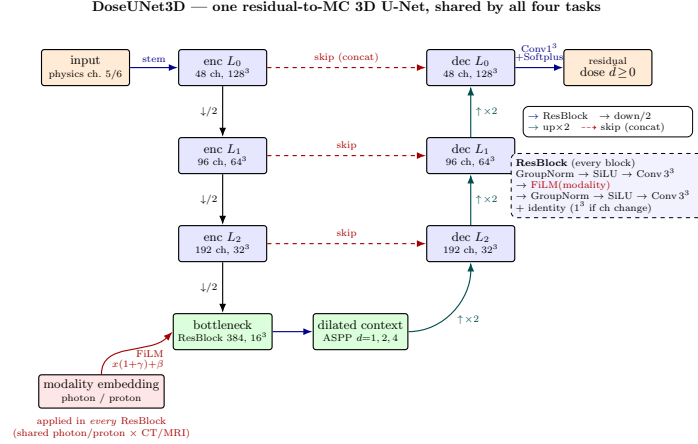


Fig. 2. The dose network $\mathcal{D}_\theta$ (DoseUNet3D): residual-to-MC 3D U-Net (channels 48/96/192/384; the deployed distilled student uses base 32), strided-conv down / transpose-conv up, dilated-context (ASPP, $d=1, 2, 4$) bottleneck for long-range lateral scatter, Softplus head (non-negative dose), FiLM modality conditioning in every ResBlock.

MC. The factorization has three measured consequences. It reduces learning to a smooth, bounded correction, so a light network trains stably; it makes the density-dependence explicit (the same operator powers our MRI-task synthesis training); and it is what makes the model robust to the benchmark’s deliberate aperture shift; training beams are *random* MLC apertures while test beams are *real clinical VMAT* apertures, and because the aperture is consumed as a fluence input rather than memorized, this shift is absorbed by construction. This report details the Photon-CT system; the same framework (shared architecture, different channels) serves our submissions to all four DoseRAD2026 tasks.

2 Methods

2.1 Data

We use only the official DoseRAD2026 training set [6] (75 patients: 39 thorax, 36 abdomen), built on the SynthRAD2025 cohort [7]: per patient a planning CT, beam parameters, and per-control-point Monte-Carlo dose-to-medium in absolute Gy (Geant4 full particle transport; the VMAT plans themselves are generated with matRad [4]) on a $2\times 2\times 2$ mm grid; 540 control points per patient. Training apertures are random; test apertures are real clinical VMAT (a deliberate distribution shift, Sec. 1). No external data are used for this task. For development we ran 5-fold cross-validation (CV) over all 75 patients (every patient held out exactly once); **the final submitted model is retrained on all 75 patients**, with the challenge leaderboard as the held-out validation signal.

2.2 Model

$\hat{d} = \mathcal{D}_\theta(\mathcal{C}(\rho, b))$, with ρ from the official piecewise-linear HU $\rightarrow$ density calibration. The operator $\mathcal{C}$ assembles **six channels** per control point, all implemented as differentiable tensor operations: (1) *density* ρ ; (2) *radiological depth*, the line integral $\int \rho d\ell$ from source to voxel, ray-marched along divergent rays with trilinear sampling; the water-equivalent depth that drives primary attenuation and build-up; (3) *primary fluence*: the MLC/jaw aperture of the control point projected through the divergent beam geometry onto each voxel and attenuated by $\exp(-\mu r \text{depth})$ with a 6 MV effective coefficient; (4) *distance to the central axis* (penumbra and off-axis softening); (5) *source distance* (inverse-square fall-off); (6) a *naive analytical dose surrogate* (fluence $\times$ inverse-square $\times$ build-up), a smooth, physically grounded starting estimate the network corrects. A per-ray *skin-entry* rule governs the depth and prior: along each ray, depth accumulates only from the first voxel with $\rho > 0.05 \text{ g/cm}^3$, and the prior is masked strictly upstream of that crossing: no dose before the body, nothing suppressed inside it (internal air cavities preserved). $\mathcal{D}_\theta$ is DoseUNet3D (Fig. 2): base 48 (teacher), four levels, ASPP bottleneck, Softplus head, FiLM conditioning; the *deployed* network is a base-32 student distilled from it (Sec. 2.3). Figure 3 visualizes the full residual-learning chain of the deployed model.

2.3 Training

Loss: high-dose-weighted L_1 to MC (high-dose fraction 0.1, weight 10) plus bounded penumbra/interface/lung terms; AdamW (lr 10^{-4} scratch, 2×10^{-5} finetune), EMA, gradient clipping; patches 128^3 , batch 2–4. The base-48 teacher trains $\sim 120\text{k}$ steps plus finetune. Two further stages define the final submission. **Multi-modal joint training:** starting from the trained photon-MRI system, the shared dose network continues training on a 50/50 interleave of real-CT cached-channel batches and MRI end-to-end steps; the resulting single network

Photon-CT (skin-entry) - 1ABB006 CP 0_134 (abdomen) | prior->GT corr 0.898 pred rel-MAE(>5%) 2.6%

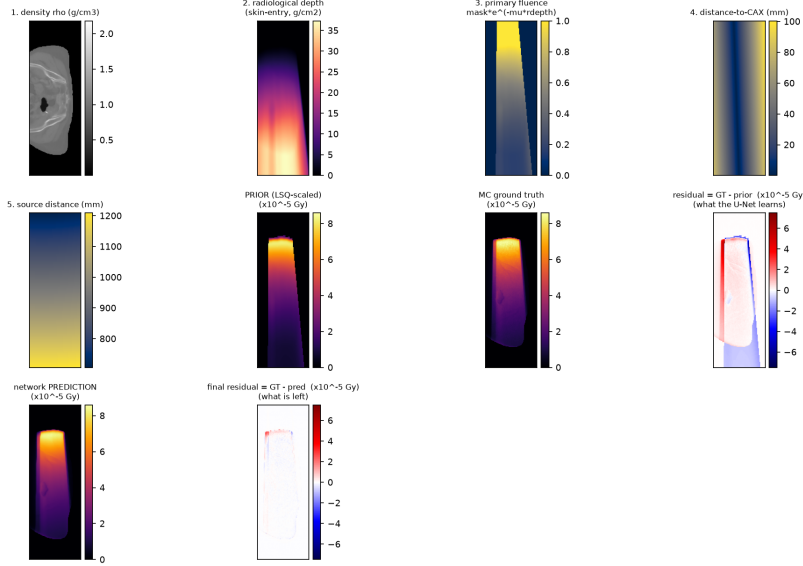


Fig. 3. Residual-learning chain of the deployed skin-entry model (abdomen, 1ABB006, peak-dose control point): input channels (density, radiological depth, primary fluence, distance-to-CAX, source distance) $\rightarrow$ LSQ-scaled naive prior $\rightarrow$ MC ground truth $\rightarrow$ residual (GT–prior) $\rightarrow$ network prediction $\rightarrow$ final residual (GT–pred). Both residual panels share one symmetric window: the shrinkage is the network’s contribution.

is a *better* CT engine than the dedicated CT model (Sec. 3.3). **Knowledge distillation (deployment):** the base-48 network is distilled into a base-32 student (7.6M vs. 17.1M parameters, $2.1\times$ faster forward) with a two-stage D-recipe: 200k steps of $L_1(\text{GT}) + 0.5 L_1(\text{teacher})$, then a 40k GT-only finetune. The student finished *above* the teacher’s held-out gamma (+0.33), so the speed is free. Reproducibility: fixed seeds, YAML-configured runs, EMA checkpoints; code and weights to be open-sourced (Sec. 6).

2.4 Evaluation

Internal: plan-level local γ [5] (pymedphys, 1%/1 mm, $\geq 10\%$ Rx cutoff, matching the official metric; verified byte-identical against the vendored official evaluation code), per-beam masked MAE, official IDD and stratified-MAE reimplementations, and a frozen 16-patient held-out cohort gating every deployment change (rule: *a change ships only if it loses neither accuracy nor speed*). A pitfall we surfaced and corrected: a photon control point has broad scatter tails ($\sim 4\%$ of plan energy outside margin-8 crops), and scoring against equally-cropped ground truth hides the loss entirely (96.2 vs. 86.4 against the full-grid GT on one patient); all crop margins were therefore selected on *full-grid* metrics. Runtime:

Table 1. Photon-CT results. CV: 5-fold over the 75 public patients. Hidden test: official metrics of the final full-data submission on the DoseRAD2026 preliminary test set (August 2026). The runtime gate is 181 s/job.

Metric	Value
CV plan $\gamma 1\%/1\text{ mm}$ (5-fold)	93.4 ± 1.0 (multi-modal variant 96.2 ± 1.1)
Hidden-test per-beam masked MAE	0.0107 ± 0.0014
Hidden-test IDD curve distance	0.0072 ± 0.0040
Hidden-test stratified plan MAE	0.0049 ± 0.0016
Hidden-test plan $\gamma 1\%/1\text{ mm}$	95.33 ± 1.89
Hidden-test DVH clinical score	0.30 ± 0.14
Hidden-test runtime per job	80.3 s (0.12 s/CP; distilled student ~ 0.09 s/CP)

the official score is $T = t_{\text{fix}} + N \cdot t_{\text{dose map}}$, gate 181 s/job, double-weighted in the ranking; we profile at job scale under the platform memory budget.

3 Results

3.1 Accuracy: cross-validated and hidden-test

Table 1 summarizes accuracy and the official hidden-test metrics. Per-fold CV $\gamma 1/1$: 94.0/92.4/94.4/94.2/92.1 (mean 93.4 ± 1.0); the multi-modal variant reaches 96.2 ± 1.1 , a +2.8 gain holding on every fold. On the fixed development fold ($n = 16$): $\gamma 1/1 = 95.7 \pm 3.8$ (abdomen 97.4, lung 94.0), $\gamma 3/3 = 99.8$; per-beam masked MAE $1.19 \pm 0.24\%$ of the beam maximum. For reference, the challenge’s analytical baseline (pyRadPlan) reaches only $\gamma 1/1 \approx 26.9$ on this metric; the residual network’s correction of the analytical prior is essential.

3.2 Ablations

The two Photon-CT-specific levers stack (single held-out fold, $n = 16$; Table 2): the naive analytical prior contributes +1.9 γ and hard-region loss weighting +2.8, together +4.0 over a plain 5-channel U-Net with uniform L_1 .

Table 2. Photon-CT ablation (plan $\gamma 1\%/1\text{ mm}$, $n = 16$): analytical prior $\times$ hard-region loss weighting.

	uniform L_1	hard-region weighting
no prior (5-ch)	91.7	94.5
+ naive prior (6-ch)	93.6	95.7

3.3 Multi-modal training exploits the photon residual’s head-room

A cross-validated *real-CT ceiling* experiment first located spare capacity: fed the real CT, the jointly trained photon-*MRI* dose network outperformed the dedicated CT network by +2.4 points (95.8 ± 1.3 vs. 93.4 ± 1.0), verified through the identical cached-channel pipeline (weights are the only difference). Testing the implied strategy directly, multi-modal joint training of one network over real- and synthetic-CT inputs raised CV Photon-CT to 96.2 ± 1.1 (+2.8, on every fold) with the MRI task unchanged. The mechanism is coherent with the residual design: the photon analytical prior is deliberately crude (it omits scatter, heterogeneity, penumbra), so the residual is large and structured, and exposure to a broader distribution of density-to-dose mappings acts as a strong, dose-relevant input augmentation. The identical experiment on the proton tasks; where the pencil-beam prior already supplies $\sim 95\%$ of the dose; is flat, a saturation control that locates the exploitable head-room precisely in the residual to a *weak* prior.

3.4 The crop margin is an accuracy–runtime knob, and cropped scoring hides its cost

We quantified a methodological pitfall that affects any cropped evaluation. A photon control point has broad scatter tails: $\sim 4\%$ of the plan’s energy falls outside margin-8 crops, concentrated in the 10–20%-of-max dose band where a *local* 1% gamma criterion has almost no tolerance. Scoring a margin-8 model against equally-cropped cached ground truth hid this entirely; 96.2 apparent vs. 86.4 true against the raw full-grid ground truth on one patient, so every margin decision was re-made on full-grid metrics. Widening the margin needs no retraining (the network is fully convolutional over analytic channels), and margin trades directly against runtime: margin 16 costs +0.01–0.02 s/CP over margin 8 but preserves the tail dose; margins 8 and 12 lost gamma and IDD on the board; a margin-8 *finetune* (network retrained at the deployed margin) did not recover the gap, showing the loss is physical tail truncation, not train–deploy mismatch. Margin 16 with cutoff-zeroed output is the deployed optimum. A related official-metric detail: the challenge’s ground truth zeroes dose below the per-beam cutoff; matching it by zeroing our sub-cutoff outputs was worth +3 γ on the photon board (for protons, quantising rather than zeroing was better), an example of why we vendored and byte-verified the official evaluation code.

3.5 Distillation: the student overtakes the teacher

The deployed network is a base-32 student (7.6M parameters) of the base-48 teacher (17.1M). Single-stage distillation converges slowly (fold-0 plan γ vs. steps: 77.7 at 40k, 86.9 at 80k, 93.2 at 160k, 94.0 at 200k = teacher parity 94.0), and the full D-recipe adds a 40k GT-only finetune, after which the full-data student *exceeded* the teacher’s held-out gamma by +0.33 at $2.1\times$ speed. Two scaling observations: a base-24 student ($2.6\times$) fell measurably short in probes, and cross-domain reuse fails; the CT-trained student loses 4.1 γ when stitched behind an

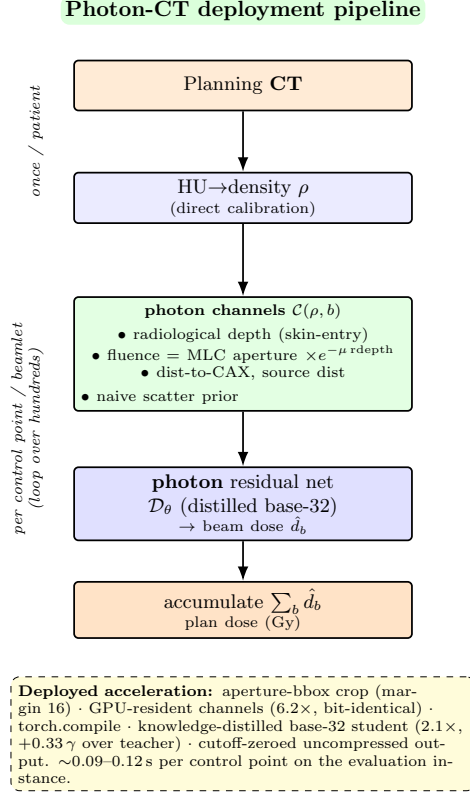


Fig. 4. The deployed Photon-CT container: per patient, the density stage runs once; the differentiable-channels $\rightarrow$ residual-net loop runs per control point/beamlet. The callout lists the deployed acceleration measures and their measured effects.

MRI front-end whose synthetic-density channels it never saw, recoverable only by finetuning on synthesis-domain channels. Distillation is thus domain-specific but, within domain, strictly dominant at this task size.

3.6 Runtime engineering

Runtime is double-weighted, so deployment was treated as a research target under the no-accuracy-loss gate. Table 3 lists the levers. Per control point, an aperture-derived bounding box (margin 16 voxels, selected on full-grid metrics) crops the compute domain; all six channels are built GPU-resident (6.2× over the CPU builder, bit-identical); the network is compiled (`torch.compile` pays off here: one grid per patient, 540 same-shape control points amortize the one-time cost); sub-cutoff outputs are zeroed and written uncompressed (photon jobs are compute-bound; compression only adds CPU). The deployed distilled stu-

Table 3. Photon-CT acceleration levers (all accuracy-gated on a frozen held-out cohort; measured, not estimated).

Lever	Effect	Accuracy impact
Aperture-bbox crop, margin 16	−30 s/job vs. margin 24	γ preserved (full-grid gated)
GPU-resident channel builder	6.2 $\times$ channel build	bit-identical
<code>torch.compile</code> (dose net)	$\sim 1.5\times$ net	exact
Distilled base-32 student (D-recipe)	2.1 $\times$ net forward	+0.33 γ over teacher
Zero-cutoff output, no compression	write path off critical path	+3 γ (cutoff match)
<i>Rejected:</i> CP batching / async emission / fp16	$\leq 1.09\times$ or negative	—

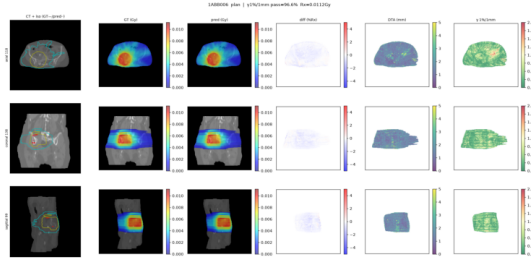
dent runs at ~ 35 ms/CP on our development GPU and ~ 0.09 – 0.12 s/CP on the evaluation instance. A negative finding worth reporting: for this small distilled network, batch-1 inference already saturates the GPU; control-point batching (1.01 – $1.09\times$), asynchronous delayed emission ($0.90\times$), fp16 weights ($1.00\times$) and autotuned compilation (—, platform t_{fix} pays the longer autotune) all failed our gate; once channel construction and the write path are off the critical path, the convolution forward itself is the deployment floor.

4 Discussion

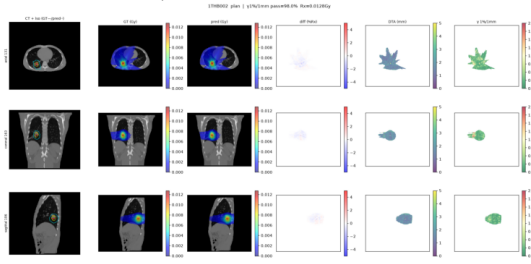
The explicit prior-plus-residual factorization, rather than raw capacity, drives this system’s accuracy, robustness and speed: the ablation shows prior and loss-weighting stack to +4.0; the aperture-as-input design absorbs the random-to-VMAT test shift; and the residual’s head-room is what multi-modal training converts into a further +2.8. The deployed model is a distilled student that is *faster and slightly better* than its teacher, evidence that at this task size the base-48 network is over-parameterized for the residual it must learn. **Limitations and generalizability.** All data come from one benchmark (abdomen/thorax, single machine model); cross-vendor and cross-site validation is future work. Ablation effects are single-fold ($n = 16$); the headline accuracy is the 5-fold CV. One geometrically hard thoracic case (large heterogeneous field) was the worst patient in all four challenge tasks simultaneously and passes clinically ($\gamma 3/3 = 88.8$ – 97.4) while failing 1%/1 mm; the residual difficulty is geometric (steep dose–anatomy gradients), not modality-specific, and argues for cross-validated reporting (Fig. 5).

Future work. Four directions follow directly from our measurements. (i) *DVH-aware selection:* our development was gamma-centric, and the DVH clinical score is the metric on which our submission ranks weakest relative to its gamma; internalizing the official DVH pipeline (structures from public segmentation models) for model selection is low-cost and likely worth more than further gamma tuning. (ii) *Student scaling:* base-32 matched-and-exceeded its teacher; mapping the accuracy–speed frontier over base-24/40 students and longer distillation schedules would establish whether another $1.3\times$ is free. (iii) *Multi-modal*

Abdomen (1ABB006) — plan $\gamma 1\%/1\text{mm} = 96.6\%$



Lung (1THB002) — plan $\gamma 1\%/1\text{mm} = 98.0\%$



Hard case (1THB143) — plan $\gamma 1\%/1\text{mm} = 68.6\%$

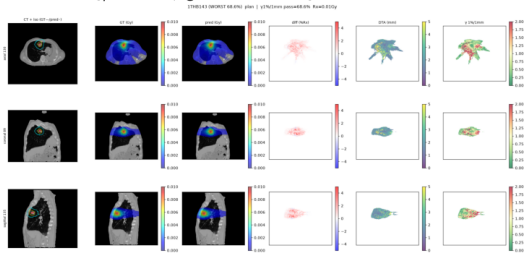


Fig. 5. Qualitative Photon-CT results on three cases spanning the difficulty range: abdomen (1ABB006, plan $\gamma 1/1 = 96.6$), lung (1THB002, 98.0), and the geometrically hard case 1THB143 (68.6; the worst patient in all four challenge tasks simultaneously, yet largely passing at 3%/3 mm: small-magnitude errors at steep lung–tissue interfaces). Columns: image with GT (solid) and predicted (dashed) isodose, MC dose, predicted dose, difference, distance-to-agreement, local $\gamma 1\%/1\text{ mm}$.

training as the default: the +2.8 cross-validated gain from joint real/ synthetic-CT training held on every fold and should be the standard recipe wherever a paired MRI task exists. (iv) *Gradient-based planning:* the engine is differentiable down to the fluence input, making it a natural physics layer for direct aperture/fluence optimization [8,9]; a proof-of-concept VMAT segment-weight optimization through the frozen engine is our next step outside the challenge.

5 Author Contributions (CRediT)

Kai Wang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Meixu Chen:** Software (deployment and runtime optimization), Investigation (acceleration study), Validation, Writing – review & editing. **Rui Yang:** Conceptualization, Methodology, Writing – review & editing, Supervision.

6 Other Information

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References

1. Nguyen, D., et al.: 3D radiotherapy dose prediction on head and neck cancer patients with a hierarchically densely connected U-net. *Phys. Med. Biol.* **64**, 065020 (2019)
2. Pastor-Serrano, O., Perkó, Z.: Millisecond speed deep learning based proton dose calculation with Monte Carlo accuracy. *Phys. Med. Biol.* **67**, 105006 (2022)
3. Hong, L., et al.: A pencil beam algorithm for proton dose calculations. *Phys. Med. Biol.* **41**, 1305–1330 (1996)
4. Wieser, H.P., et al.: Development of the open-source dose calculation and optimization toolkit matRad. *Med. Phys.* **44**, 2556–2568 (2017)
5. Low, D.A., Harms, W.B., Mutic, S., Purdy, J.A.: A technique for the quantitative evaluation of dose distributions. *Med. Phys.* **25**, 656–661 (1998)
6. DoseRAD2026 Grand Challenge, <https://doserad2026.grand-challenge.org> (2026)
7. Thummerer, A., et al.: SynthRAD2025 Grand Challenge dataset: generating synthetic CTs for radiotherapy. *Med. Phys.* (2025)
8. Simkó, A., et al.: A physics-informed, plug-and-play dose engine for gradient-based radiotherapy treatment planning. *arXiv:2512.18863* (2025)
9. Achlatis, S., Gavves, E., Sonke, J.-J.: Physics-guided radiotherapy treatment planning with deep learning. *arXiv:2506.19880* (2025)